Algebra 1 Name____________________
Unit 3 Assessment Date________________
Calculator Period_________

1. Noah placed an empty bowl on a scale. He added different amounts of a liquid to the bowl, recording the amount of liquid added, in tablespoons, and the total weight of the bowl and liquid, in grams, each time on a scatter plot. He fit the line $y = 14.75x + 861.82$ to the data in his scatter plot, where x represents the amount of the liquid, in tablespoons, and y represents the total weight of the bowl and the liquid, in grams.

a. Interpret the slope of the line based on the context.

b. Interpret the y-intercept of the line based on the context.

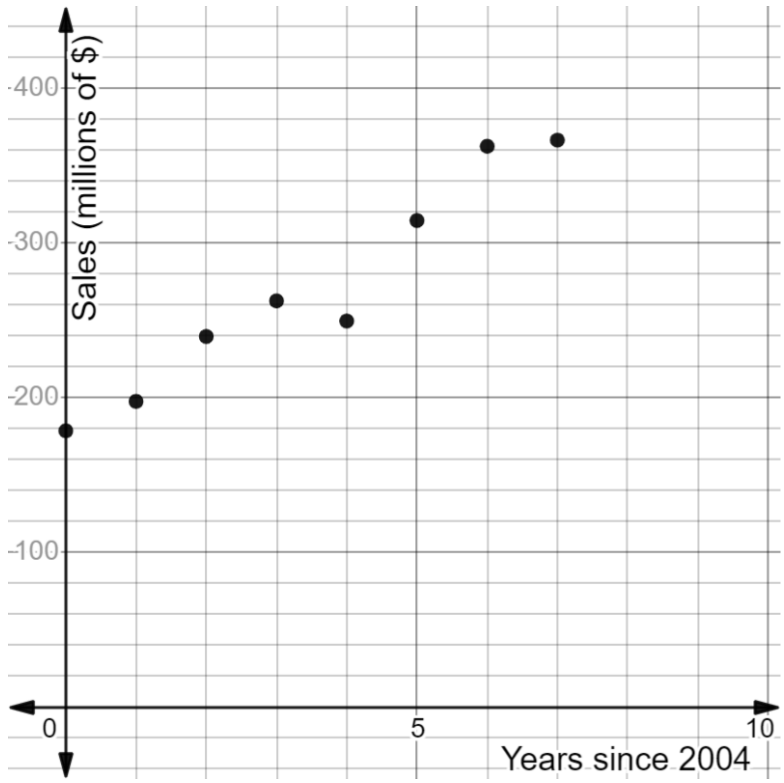
c. Determine the weight, in grams, when 18 tablespoons of the liquid are placed in the bowl.

Weight (grams)	

d. Determine the approximate number of tablespoons of the liquid Noah needs to add so that the total weight is $1{,}200$ grams.

Amount of Liquid (tablespoons)	

2. The total sales, in millions of dollars, of Florida strawberries for the years 2004 – 2011 is shown on the coordinate grid.



a. Draw a trend line on the scatter plot.

b. Use your trend line to write an equation that would fit the data.

Equation	

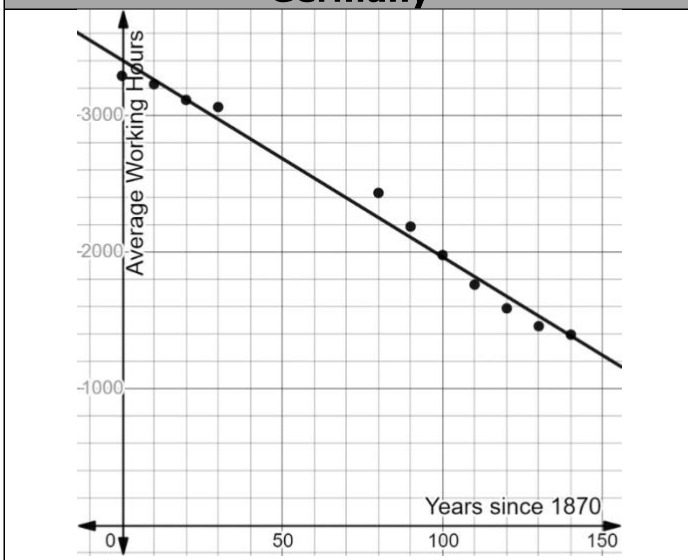
c. Interpret the slope of the line based on the context.

d. Interpret the y-intercept of the line based on the context.

3. A social scientist researched how the average working hours per worker over a full year for two different European countries changed since 1870. She created a scatter plot and found the equation for the line of best fit for each set of data. She then used this equation to create a table of values for the residuals.

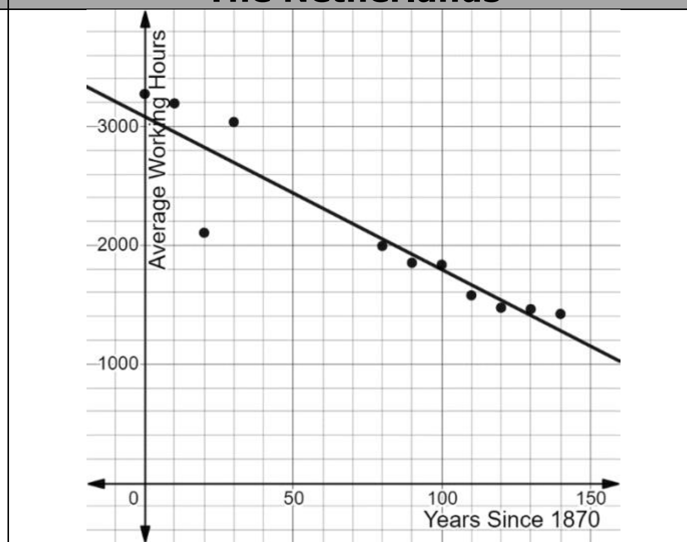
The scatter plots, equations for the lines of best fit, and residuals for the two countries are shown, where x represents the number of years since 1870 and y represents the average working hours per worker over a full year.

Germany



$y = -14.4166x + 3399.86$

The Netherlands



$y = -12.8708x + 3082.74$

Years since 1870	Residuals
0	-115.86
10	-32.694
20	-3.528
30	88.638
80	180.8967
90	78.7563
100	14.3177
110	-57.9257
120	-87.2645
130	-73.6891
140	8.3469

Years since 1870	Residuals
0	191.26
10	239.968
20	279.676
30	340.384
80	-59.0811
90	-72.1563
100	39.9225
110	-88.5915
120	-64.3725
130	52.4852
140	140.5132

a. Complete the statement.

The data for The Netherlands has a
- ○ positive
- ○ negative

correlation because

- ○ the slope of the line of best fit is
- ○ most of the residual values are

- ○ positive.
- ○ negative.

b. Complete the statement.

○ Germany ○ The Netherlands	has a	○ weaker ○ stronger	correlation than	○ Germany ○ The Netherlands

because the residuals have a
○ larger
○ smaller
absolute value, indicating that the

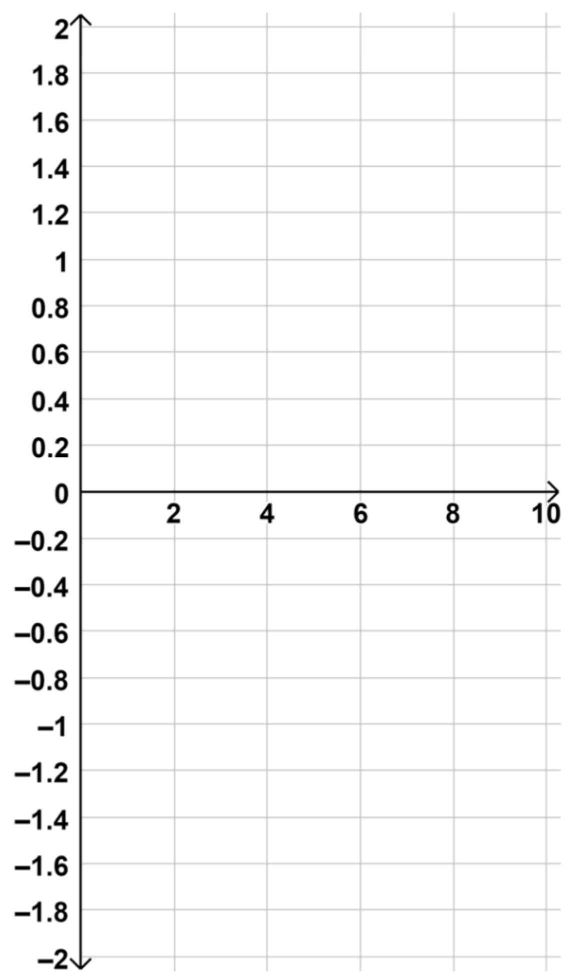
distance each
○ actual
○ predicted
value is from the line of best fit is
○ larger.
○ smaller.

4. Each year, the State of Florida collects data on the price per box for each type of orange. A mathematician for a grower's association analyzed the data for the years 2009 to 2018 for a certain variety of orange. Some of the data is shown.

Years Since 2009	0	1	2	3	4	5	6	7
Price ($)	5.95	7.11	8.88	6.25	8.41	8.40	8.99	10.50

The mathematician found the equation $y = 0.5x + 6.31$ models the data.

a. Plot the residuals.



b. Explain whether or not the linear function is a good fit for this data set.